Analysis of the Risk of Natural Disasters

Written Report

CIVE 202-152

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Introduction

The National Risk Index (NRI) is a measurement conducted by the Department of Homeland Security for the Federal Emergency Management Agency (FEMA) to determine the relative risk of natural disaster events in the United States. FEMA classifies risk as the product of probability of an event times the impact of such an event occurring. With the use of the National Risk Index Relative Risk Index percentiles and ratings are produced for further analysis of the Expected Annual Loss due to natural disasters, Social Vulnerability in differing regions and the community's ability to be resilient after experiencing natural disasters. Each component also has their own percentiles and ratings that can be viewed as a composite score for all hazards or individually for each of the 18 Hazards FEMA identifies such as Avalanches, Coastal Floodings, Cold Waves, Droughts, Earthquakes, Hail, Heat Waves, Hurricanes, Ice Storms, Inland Floodings, Landslides, Lightning, Strong Winds, Tornados, Tsunamis, Volcanic Activity, Wildfires, and Winter Weather. For this project, our team has been hired by Risk Averse, LLC. An independent risk analysis consulting research group. To perform a sensitivity analysis, it's the process of changing or altering the inputs to a model or methods and determining how great a change our outputs have when compared to the controlled model or method. In this project, it will be FEMA's National Risk Index and our own version of observations and chosen hazards.

Project Goals

The goal of this project is to evaluate potential bias in the FEMA National Risk Index methodology by conducting a sensitivity analysis and developing an alternative definition of risk. This analysis will be performed using Census Tract-level data for two selected states: **Iowa and California**. These states were selected to provide contrast between a predominantly rural state and a highly urbanized state, allowing for meaningful comparison of risk distribution. Specifically, this project aims to understand and document how risk is defined and calculated in the NRI, develop an alternative risk index using external research and datasets, compare NRI results with the proposed model to identify potential categorical bias, and communicate findings through data visualization, summary tables, and geospatial mapping.

The National Risk Index defines **risk** as the combination of three core components: **Expected Annual Loss (EAL), Social Vulnerability (SVI), and Community Resilience (CRI).**

Project Tasks:

Task 1: Data Organizing and Cleaning

Our group will download all the data for Nebraska as well as find and download the data for the two other states we will be researching. Raw datasets will be cleaned using Python. This includes handling missing values, renaming variables, formatting data types, and ensuring consistency across datasets. Data will be merged using common geographic identifiers to create a unified dataset for analysis.

Task 2: Research and Development

The team will research and document how FEMA defines and calculates risk, including Expected Annual Loss, Social Vulnerability, and Community Resilience. The original NRI risk formula and percentile-based ranking system will be replicated using Python to ensure a baseline for comparison. Using external research and engineering judgment, the team will develop an alternative risk model. This model may adjust the weighting of EAL, SVI, and CRI or incorporate additional considerations to better reflect community impacts. The purpose is to test how sensitive risk outcomes are to changes in the definition of risk.

Task 3: Sensitivity Analysis and Comparison

A sensitivity analysis will be performed by comparing outputs from the NRI model and the proposed model. Differences in percentile rankings and categorical ratings will be analyzed to identify potential bias in how risk is currently defined.

Task 4: Visualization and Mapping

The team will generate at least two summary tables or figures comparing risk results between models. Visualizations will include statistical summaries (mean, max, distribution) and comparative plots. Additionally, a geospatial map will be created using GeoPandas to visualize the spatial distribution of risk across both states.

Task 5: Interpretation and Report

All results will be interpreted with attention to ethical implications, including how differences in risk definitions may impact funding allocation, disaster preparedness, and community decision making. This will be written in a report that examines what implications these differences will have on the people impacted by natural disasters and how this relates to being able to give communities FEMA funding in the future to mitigate disasters.

Methods

1. Data Collection and Integration

   NRI Census Tract-Level data and the CDC Social Vulnerability Index data were collected and merged using FIPS codes. Metadata files were used to interpret variable definitions and ensure consistency.

2. Data Cleaning and Preprocessing

   Missing and placeholder values (e.g., -999, -9999) were converted to NaN and handled appropriately. Variables were normalized where necessary to allow comparison across datasets

3. Risk Model Development

   The NRI risk model was based on the Expected Annual Loss (EAL), Social Vulnerability (SVI), and Community Resilience. While the Alternative Risk Model was developed by adjusting the weight of key variables and emphasizing social vulnerability impacts.

4. Sensitivity Analysis

Both models were compared by evaluating differences in percentile ranking and categorial classifications. This allowed identification of potential boas in how risk is assigned across regions.

Results

After completing the Python code for this project. We observed that the alternative risk model produced noticeable shifts in risk rankings, particularly in areas with high social vulnerability. Urban regions for California tended to show higher vulnerability in risk depending on the model used. Rural regions for Iowa showed more stable risk classifications across models. The

difference between models highlights how significant weighting choices influenced the perceived risk.

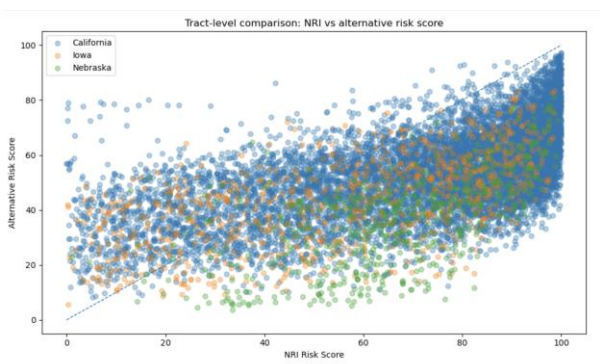
NRI category counts

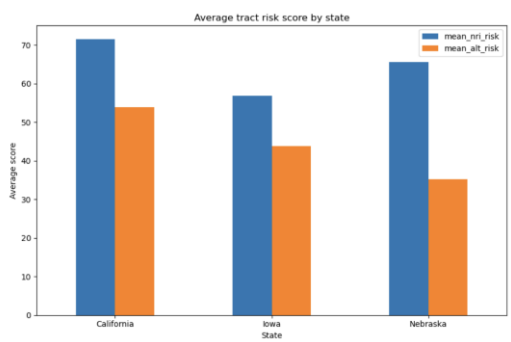
RISK_RATNG	Relatively High	Relatively Low	Relatively Moderate	Very High	Very Low
STATE					
California	2938	1750	2967	913	537
Iowa	156	286	341	5	108
Nebraska	84	210	232	11	16

Alternative category counts

ALT_RISK_RATNG	Relatively High	Relatively Low	Relatively Moderate	Very High	Very Low
STATE					
California	2688	1625	4182	488	123
Iowa	134	278	398	7	79
Nebraska	46	184	195	1	127

Our summary tables and visualizations demonstrated the distribution of risk scores across states, the comparison of mean and the maximum risk values, as well as the spatial variation in risk using geospatial mapping. To conclude, this project demonstrates that risk classification nis highly sensitive to how risk is defined and calculated. The comparison between the NRI methodology and the alternate model reveals that risk isn't purely objective and can be influenced by model assumptions. Social vulnerability plays a critical role in shaping risk outcomes, and the use of different definitions of risk can lead to different policies and funding decisions.





Conclusion

This project demonstrates that the definition of risk plays a critical role in determining outcomes in natural disaster analysis. The sensitivity analysis reveals that risk classifications are highly dependent on model assumptions. Also, the NRI methodology may introduce bias by emphasizing economic loss. Alternative definitions that prioritize social vulnerability can significantly change risk rankings. These findings highlight the importance of carefully considering how risk is defined, especially when it influences funding, policy decisions, and community resilience strategies. Future work could expand this analysis to additional states,

incorporate more variables, and further refine weighting strategies to better balance economic and social considerations.

References

Janssen, C. (2026). *CIVE202 Spring 2026 Project 2 Python Code Notebook.* Submitted as part of course deliverables.

"National Risk Index Data." *Fema.gov*, 6 Jan. 2026, www.fema.gov/about/openfema/data-sets/national-risk-index-data.